

RAT ground stow

The ram air turbine (RAT) of both aircraft can be stowed again after the landing, on the ground only, as the FCOMs describe; a request in flight or while an extension is still commanded is refused

Aircraft	A32NX and A380X
Where to find it	The aircraft systems (A32NX hydraulic RAT, A380X electrical RAT); requested from the flyPad Ground > Services page (row on <code>core/efb-redesign</code>)
Branch	<code>core/rat-ground-stow</code> , based on <code>master</code> (FlyByWire <code>master 255546cb0</code>); independent of the other core branches. It holds the systems part (shared Rust, tests, <code>simvar docs</code> , <code>CHANGELOG</code> line). The flyPad row that sends the request is on <code>core/efb-redesign</code> , because it is a row of the Material 3 Services page: the feature works for the user only when both branches are merged. Without the flyPad row the request can be written as <code>L:A32NX_RAT_STOW_REQUEST</code> with any <code>L:var</code> tool; without this branch the flyPad row shows while the RAT is out but pressing it has no effect.
Status	Sim-tested by the user on both aircraft on develop: OK (2026-10-05). Not sim-tested on the master base; Rust tests pass there.
Pictures	No pictures: the flyPad row is one line of the Services equipment list; the A380X RAT rig and the A32NX HYD page are stock.

1. RAT stow on the ground

Aircraft: A32NX and A380X Where: flyPad > Ground > Services, row "RAT stow (maintenance)" Status: Tested in the sim by the user on develop: OK on both aircraft (2026-10-05). Rust tests in systems (<code>ram_air_turbine_deployment.rs</code>), <code>a320_systems</code> (hydraulic) and <code>a380_systems</code> (electrical); <code>vitest ratStow.spec</code> on <code>core/efb-redesign</code> .

What it is for

Once the RAT had extended (both AC buses lost, RAT MAN ON, A32NX EMER ELEC PWR MAN ON), nothing stowed it again: it stayed out for the rest of the session, and on the A32NX HYD RAT FAULT stayed on. On the real aircraft the ground crew stows it after the flight.

NOTE A320 FCOM DSC-29-10-20 (<code>a320_fcom.txt</code> l.43637-43653): "The RAT deploys automatically if AC BUS 1 and AC BUS 2 are both lost. It can be deployed manually from the overhead panel. It can be stowed only when the aircraft is on the ground."

NOTE A380 FCOM DSC-29-10 (<code>a380_fcom.txt</code> l.61250): "The green hydraulic system powers RAT retraction, on ground."
--

The systems now accept a maintenance stow request (`L:A32NX_RAT_STOW_REQUEST`) on the ground, while no extension is commanded. One shared Rust model (`ram_air_turbine_deployment.rs`) moves the strut of both aircraft, so the A320 and the A380X behave the same.

What you should see

Situation	What you should see
RAT stowed	No RAT row in the Services list.
RAT out, aircraft on the ground and stopped	Row "RAT stow (maintenance)", status "RAT extended", button Stow .
Stow pressed and accepted	Status "Stowing" for about 3 s; the strut moves in 1 s; the row disappears. A380X: the RAT rig retracts. A32NX: the HYD page RAT indication goes back to stowed.

RAT out in flight, or on the ground but moving	Row greyed, status "RAT extended · stow when stopped on the ground".
Stow pressed while an extension is still commanded (emergency electrical configuration, RAT MAN ON or A32NX EMER ELEC PWR MAN ON held in)	Refused: the RAT stays out.
RAT MAN ON (or the automatic extension) after a stow	The RAT extends again.
A32NX on the ground with the RAT out	HYD RAT FAULT, as the FCOM says (PRO-ABN-HYD, a320_fcom.txt l.88843-88851: "The RAT is not fully stowed"); it clears after the stow.

Limitations and why

Limitation	Why	What to do
The stow needs no hydraulic pressure.	Design choice: the FCOMs say only "on the ground" (A320) and that green hydraulics power the retraction on ground (A380); the stow is a ground crew action, so it is not tied to the aircraft hydraulics being pressurised.	-
The strut travels in 1 s.	Design choice: the same speed as the stock extension; no FCOM value.	-
No RAT on the A32NX model.	The A32NX exterior model has no RAT yet: only the HYD page, HYD RAT FAULT and the RAT pump show the stow.	-
A request in flight is dropped.	Each request is evaluated once (written back to 0), so a request in flight does not stow the RAT after the landing.	Request again on the ground.
The flyPad row needs the stationary aircraft.	Design choice of the flyPad (like the other ground services); the systems themselves only need "on the ground".	Stop the aircraft.

How to use

1. After a flight with the RAT out, stop on the ground.
2. Restore normal electrical power (engines or APU generators, or external power) so that no extension is commanded; on the A32NX the RAT MAN ON and EMER ELEC PWR MAN ON pbs are not pressed.
3. flyPad > Ground > Services: press **Stow** on the row "RAT stow (maintenance)".

How to test

The flyPad row is on `core/efb-redesign`: test with both branches merged. With this branch alone, replace each "press Stow" step by setting `L:A32NX_RAT_STOW_REQUEST` to 1 with an L:var tool (SimConnect); the "row" lines then do not apply.

#	Do this	Expected result
1	A380X on the ground, engines running, parking brake set. Overhead EMER ELEC PWR panel: open the guard and press RAT MAN ON.	The RAT extends (rig out); the flyPad Services list shows "RAT stow (maintenance)", active.

2	With RAT MAN ON released, press Stow on the flyPad.	"Stowing", the rig retracts in about 1 s and the row disappears. (While RAT MAN ON is held in, the extension is still commanded and the stow is refused.)
3	A380X in flight: press RAT MAN ON, then look at the flyPad.	Row greyed ("stow when stopped on the ground"); a request written in flight is dropped.
4	Land with the RAT out, stop, press Stow.	The RAT stows.
5	A32NX on the ground, engines running. HYD panel: RAT MAN ON (or EMER ELEC PWR MAN ON).	The RAT extends: HYD page RAT out, HYD RAT FAULT (FCOM: the RAT is not fully stowed); flyPad row "RAT stow (maintenance)".
6	Press Stow on the flyPad.	The HYD page RAT goes back to stowed, HYD RAT FAULT clears, the row disappears.
7	A32NX: press RAT MAN ON again.	The RAT extends again.